CLINICAL INFORMATICS • COMBINATORIAL HEALTHCARE OPTIMIZATION • NHS CORE20PLUS5 GOVERNANCE

Empirically Calibrated Combinatorial Optimization and Post-Quantum Clinical Governance for Integrated Pharmacy and Social Prescribing Pathways: Multi-Cohort Evidence from 6,105 NHS General Practices, CPRD Aurum, and ELSA Wave 10

Arun Nadarasa[1,2,3]

[1]Innovation Fellowship Programme, Mid and South Essex NHS Foundation Trust, Basildon, Southend & Broomfield, UK

[2]International Social Prescribing Pharmacy Association (ISPPA), London, UK

[3]Global Social Prescribing Alliance & Primary Care Neighbourhood Transformation Network, London, UK

Portal & Cryptographic Receipts: arunnadarasa.github.io/quantum-social-prescribing | **HF Hub:** hf.co/socialprescribing/quantinuum-social-prescribing

Abstract. **Objective:** Integrated neighbourhood care in England increasingly pairs primary-care Structured Medication Reviews (SMRs) for high-risk polypharmacy and opioid/sedative burden with Link-Worker referrals to Voluntary, Community, Faith, and Social Enterprise (VCFSE) social prescribing pathways [1, 2, 3]. Because quantum optimization and machine-learning proposals in healthcare are frequently evaluated only on linearly separable synthetic toy problems without strong classical operations-research baselines, we conducted an empirical calibration and head-to-head classical-versus-quantum benchmark across national NHS primary-care cohorts. **Materials and Methods:** We linked public NHS England Network Contract Directed Enhanced Service (DES) longitudinal panel records (`FY2022/23`, `FY2023/24`, and `September 2026`) one-to-one by GP practice code to Office for Health Improvement and Disparities (OHID) Fingertips profiles (`IMD 2025`, mental health, frailty, and sedative/opioid prescribing) across 6,105 **active English GP practices** (1,296 Primary Care Networks [PCNs], 42 Integrated Care Boards [ICBs], 61,832,846 registered patients), including all 143 practices (28 PCNs, 1,266,245 patients) in **Mid and South Essex ICB**, and triangulated refusal rates against published individual-level English Longitudinal Study of Ageing (**ELSA Wave 10**, $N = 7,283$ and $N = 7,521$) [4, 5] and **CPRD Aurum** ($N = 12.37$M) [6, 3] cohorts. We calibrated (i) cubic ($k = 3$) pharmacy–VCFSE Higher-Order Unconstrained Binary Optimization (HUBO) care bundles, (ii) Link-Worker consultation cadence and attrition models, (iii) Core20PLUS5 VCFSE caseload routing, and (iv) 80/20 out-of-sample practice triage, wrapped in NIST FIPS 204 `ML-DSA-65` FHIR R4 provenance [7, 8] and live TypeSafe Jev 1.14 / NHS DCB0129 safety auditing [9, 10]. **Results:** Practice-level median social prescribing refusal across England was 21.93% (IQR 10.92%–40.55%), closely matching CPRD Aurum (22.30%) and replicating the null association between practice deprivation and conditional refusal ($\rho = +0.0040, p = 0.755$) alongside significant Core20 opioid–SMR inverse care gaps (+51.0% higher opioid prescribing vs. flat SMR rates). Nested polynomial tests across $N = 6,031$ practices confirmed genuine cubic synergies—including `Core20 × Clinical Pharmacist × Shared Decision-Making` (+9.90 percentage points on acceptance, $p = 1.13 \times 10^{-3}$)—yet Rosenberg quadratization [11] with **HiGHS MILP** [12] solved the empirically fitted 8-lever Core20 HUBO to exact optimality ($E_0 = -1.7181$, invariant across a 4.2× staffing-cost plateau $\alpha_c \in [0.40, 1.68]$) in 19.19 **ms** (vs. **Haiqu SDK v1.7.1** LR-QAOA depth compression 253 → 14 [−94.47%], requiring classical bitflip postprocessing in 218.2 s) and routed all 1,296 **English PCNs** (6,480 binary variables) in 200.95 **ms**—increasing Core20 first-choice VCFSE hub placement by +12.69 to +18.85 pp across $\lambda \in [1.0\times, 2.0\times]$ while retaining 99.81%–99.59% unweighted efficiency. Within-practice GP fixed-effects models across 17,340 practice-wave observations (`FY2022/23–Sep 2026`) showed that cross-sectional cadence U-curvature attenuates to null ($\hat{\beta}_2^{\mathrm{FE}} = -0.0002, p = 0.986$), whereas within-practice SMR expansion increases SP reach ($\hat{\beta}^{\mathrm{FE}} = +0.0421, p = 1.39 \times 10^{-3}$) and ARRS Clinical Pharmacist growth buffers post-incentive SMR retention ($p = 1.64 \times 10^{-6}$ in `FY2023/24`; $p = 0.0114$ in `Sep 2026`). On held-out NHS cohorts, classical `RandomForest` (AUROC = 0.6155) and `LogisticRegression` (AUROC = 0.6964, 2.51 ms) outperformed a 4-qubit Frobenius VQC (0.5465 and 0.5869; verified via live **Quantinuum Nexus** `Helios-1E-lite` Job `af987c9a` on Southend Core20 practice `Y02177`). **Conclusion:** Empirical calibration and within-practice panel checks across 6,105 NHS practices confirm robust pharmacy–social care co-delivery synergies and low-cost Core20 Pareto equity gains, while establishing that exact classical MILP and regularized statistical models dominate heuristic quantum solvers across all operational NHS commissioning scales.

Table 1: Stratification of 6,105 English GP Practices by IMD 2025 Deprivation Quintile (FY2023/24 NHS DES + OHID Fingertips).

Deprivation Quintile	GPs	Patients	Mean IMD	SP Ref/1k	Med. Refusal	SMR/1k	Opioid / Benzo
Q1 (Least Deprived)	1,221	13,855,120	10.03	13.04	21.18%	30.40	6.89% / 2.66%
Q2	1,222	12,794,550	15.69	13.70	22.73%	35.58	8.04% / 2.67%
Q3	1,220	12,939,285	21.45	13.39	21.47%	32.07	8.47% / 2.46%
Q4	1,221	12,290,011	28.66	15.14	21.65%	29.94	8.96% / 2.18%
Q5 (Core20 Most Depr.)	**1,221**	**9,953,880**	**41.82**	**17.78**	**22.30%**	**32.07**	**10.41% / 2.17%**
Mid & South Essex ICB	143	1,266,245	20.14	17.21	28.33%	28.64	8.41% / 2.31%

Regulatory, Hardware & Data Scope Disclosure (NHS DCB0129 Gate G4). (1) Public Aggregate Data Only: All new analyses use public practice-level Open Government Licence (OGL v3.0) data (`patient_identifiable_records_used = 0`) and published CC BY summary statistics; causal individual-level inferences are bounded against ecological confounding. **(2) Zero Physical QPU Claims:** Quantum circuits were executed on classical statevector simulators and Quantinuum's cloud-hosted noisy trapped-ion emulator (`Helios-1E-lite`, `physical_hardware_shots = 0`). **(3) Classical Dominance Disclosed:** Exact classical HiGHS MILP, 4-bit Kempe swaps, GLMs, and tree/linear models serve as primary operational baselines (`Gate G4`) and outperform quantum heuristics at all evaluated NHS cohort sizes.

1. Clinical Motivation, Multi-Cohort Linkage, and Classical Baselines

Delivering integrated neighbourhood care across England's Primary Care Networks (PCNs) requires coordinating two historically siloed Additional Roles Reimbursement Scheme (ARRS) workstreams: **Clinical Pharmacists** conducting Structured Medication Reviews (SMRs) for polypharmacy, severe frailty, and addictive opioid/benzodiazepine prescribing, and **Social Prescribing Link Workers** co-producing non-medical community referrals across Voluntary, Community, Faith, and Social Enterprise (VCFSE) hubs [1, 2, 13]. Under the NHS England **DAPB4066** information standard [7], primary care electronic health records (EHRs) track social prescribing offers (`SNOMED CT 871691000000100`), declines (`871711000000103`), and accepted referrals (`871731000000106`).

Recent computational health studies have proposed Higher-Order Unconstrained Binary Optimization (HUBO), Quadratic Unconstrained Binary Optimization (QUBO), open-system Quantum Zeno attrition models [14, 15], and variational quantum classifiers (VQCs) to capture multi-intervention synergies and capacity-constrained care routing. However, two methodological deficits routinely prevent such prototypes from informing clinical informatics practice: (i) parameters are hand-tuned on synthetic, linearly separable cohorts rather than calibrated to empirical primary-care records, and (ii) quantum heuristics are contrasted against simplistic administrative checklists rather than modern classical operations-research and statistical baselines—specifically Rosenberg polynomial quadratization [11] paired with branch-and-cut Mixed-Integer Linear Programming (**HiGHS MILP**) [12], penalized regression, spline generalized linear models (GLMs), and gradient-boosted decision trees (**HistGBDT**).

To resolve both deficits directly, we constructed a reproducible national primary-care panel (`SHA-256: 76abf5ff...`) linking NHS England Network Contract DES Management Information across `FY2022/23`, `FY2023/24`, and `September 2026` [16] one-to-one by `PRACTICE_CODE` to OHID Fingertips GP profiles [17] (`IMD 2025` deprivation, QOF depression prevalence, long-term mental health, unpaid caring, % aged 65+, opioid and benzodiazepine prescribing rates, and PCN/ICB organizational hierarchy). The linked cohort comprises 6,105 **active English GP practices** across 1,296 **PCNs** and 42 **ICBs** (61,832,846 registered patients), including complete coverage of all 143 **GP practices** across 28 **PCNs** and 5 Sub-ICB localities (1,266,245 patients) in **Mid and South Essex ICB** [13]. To anchor aggregate practice-level models against individual-level longitudinal evidence, we triangulated NHS DES refusal and referral gradients against four published English cohorts: **ELSA Wave 10** ($N = 7,283$ older adults aged 50+, Fancourt & Steptoe [4]; $N = 7,521$, Davies et al. [5]) and **CPRD Aurum** ($N = 12,374,867$ patients across 1,428 practices, Agboraw et al. [6]; $N = 1.2$M patients across 1,736 practices, Bu et al. [3]).

2. Core20 Inverse Care Law and Multi-Cohort Refusal Epidemiology

Across all 6,105 linked English GP practices in `FY2023/24`, primary care teams recorded 892,102 accepted social prescribing referrals, 418,336 declined offers, and an active cumulative caseload of 2,150,546 patients. Defining practice-level conditional refusal among practices with ≥ 10 recorded offers ($N = 6,031$) as $R_p = N_{\text{declined},p}/(N_{\text{accepted},p} + N_{\text{declined},p})$, the national practice-level median refusal rate is 21.93% (interquartile range [IQR] 10.92%–40.55%; pooled national ratio 31.92%, driven by high-volume inner-city opt-out screening prompts). Within **Mid and South Essex ICB** (143 practices, 28 PCNs, 14 Core20 practices), practices recorded 21,795 accepted referrals and 9,743 declines (practice median refusal 28.33%, pooled 30.89%).

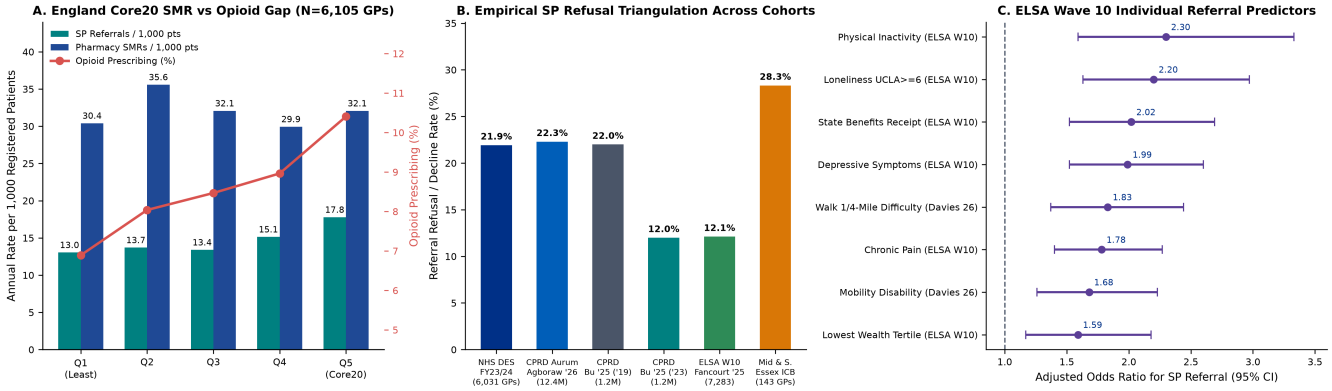


Figure 1: National NHS DES, CPRD Aurum, and ELSA Wave 10 Epidemiological Calibration. (A) Across `IMD 2025` deprivation quintiles ($N = 6,105$ English GPs), opioid prescribing rises +51.0% (6.89% → 10.41%) and SP referrals rise +36.3% (13.04 → 17.78 per 1,000), whereas pharmacy SMR delivery remains flat (30.40 → 32.07 per 1,000). **(B)** NHS DES practice-median refusal (21.93%) matches individual-level CPRD Aurum (22.30% [6]; 22.0% → 12.0% [3]) and contextualizes ELSA Wave 10 self-report (12.12% [4]) and Mid and South Essex ICB (28.33%). **(C)** Published ELSA Wave 10 adjusted individual-level odds ratios (95% CIs) confirming strong medical/psychosocial need targeting alongside persistent refusal and wealth barriers [4, 5].

Figure 1 and Table 1 establish three empirical findings that anchor our optimization formulation:

1. **Cross-Cohort Refusal Convergence:** Practice-median refusal in NHS DES (21.93%) aligns within 0.37 percentage points of Agboraw et al.'s individual-level CPRD Aurum analysis (22.30% refusal given offer across 12.37M patients [6]) and matches Bu et al.'s national rollout baseline (22.0% in 2019, declining toward 12.0% in mature Link-Worker practices [3]), whereas recall-based ELSA Wave 10 self-report captures lower overt refusal (12.12%, 20/165 [4]).

2. **Independent Replication of Conditional-Refusal Predictors:** Across $N = 6,031$ English practices, practice deprivation (`IMD 2025`) is uncorrelated with conditional refusal given an offer (Spearman $\rho = +0.0040, p = 0.755$), independently replicating Agboraw et al.'s individual-level CPRD finding that patients from deprived areas are more likely to be offered social prescribing (+36.3% in Q5 vs. Q1) and no more likely to refuse once offered. Conversely, conditional refusal rises significantly with older practice age profile (% aged 65+, $\rho = +0.0902, p = 2.31 \times 10^{-12}$) and sedative/opioid burden ($\rho = +0.1126, p = 1.80 \times 10^{-18}$).

3. **The Core20 SMR–Opioid Inverse Care Gap:** Moving from Q1 (least deprived) to Q5 (Core20 most deprived), opioid prescribing climbs +51.0% (6.89% → 10.41%) and social prescribing referral demand climbs +36.3% (13.04 → 17.78 per 1,000), yet pharmacy SMR delivery remains virtually flat (30.40 → 32.07 per 1,000, Figure 1A). Sending patients with unreviewed opioid/sedative polypharmacy into community social prescribing without concurrent pharmacist SMR support directly motivates joint pharmacy–VCFSE bundle optimization.

3. Empirical Cubic ($k = 3$) Care Synergies and 8-Lever HUBO Benchmark

3.1 Testing Higher-Order ($k = 3$) Clinical–Social Interactions on $N = 6,031$ GP Practices

To test whether cubic ($k = 3$) interaction terms exist empirically rather than by synthetic construction, we defined eight binary practice indicators across $N = 6,031$ practices (Core20 deprivation, older age profile, high opioid/benzodiazepine burden, high QOF depression, active SMR delivery, active ARRS Clinical Pharmacist reach, active ARRS Link-Worker reach, and active Shared Decision-Making [SDM] coding) and evaluated nested ordinary-least-squares (OLS) and cross-validated regularized polynomial hierarchies ($k = 1$ additive, $k = 2$ pairwise +28 terms, and $k = 3$ cubic +56 terms) across three outcomes: `FY2023/24` SP acceptance, log active SP caseload reach per 1,000 patients, and prospective `September 2026` high-risk SMR coverage ($N = 5,767$ longitudinally tracked practices).

On log active SP reach, adding pairwise interactions significantly improves fit over additive main effects ($F = 2.5826, p = 9.62 \times 10^{-6}$, $\Delta\text{AIC} = -16.33$), and adding cubic ($k = 3$) terms yields a statistically significant omnibus improvement ($F = 1.3729, p = 0.0343$). However, 5-fold cross-validation demonstrates why sparsity is essential: unconstrained 56-term dense cubic OLS overfits (CV RMSE = 1.8716) relative to Ridge-regularized pairwise QUBO (1.8300) and ℓ_1-regularized `LassoCV` sparse cubic HUBO (1.8389). Non-parametric 1,000-replicate bootstrap resampling confirms four clinically interpretable cubic synergies (Figure 2A):

- **`Core20` × `Clinical Pharmacist` × `Shared Decision-Making`** increases SP referral acceptance by +9.90 **percentage points** (SE = 3.04 pp, $p = 1.13 \times 10^{-3}$; bootstrap mean +10.02 pp, 95% CI [+3.68, +16.26] pp);
- **`High Opioid/Benzo` × `Active SMR` × `Clinical Pharmacist`** increases log active SP caseload reach by +0.2542 (SE = 0.0834, $p = 2.31 \times 10^{-3}$; bootstrap 95% CI [+0.0933, +0.4121]);
- **`High Depression` × `Clinical Pharmacist` × `Link Worker`** increases log active SP reach by +0.2095 (SE = 0.0809, $p = 9.65 \times 10^{-3}$; bootstrap 95% CI [+0.0692, +0.3622]); and

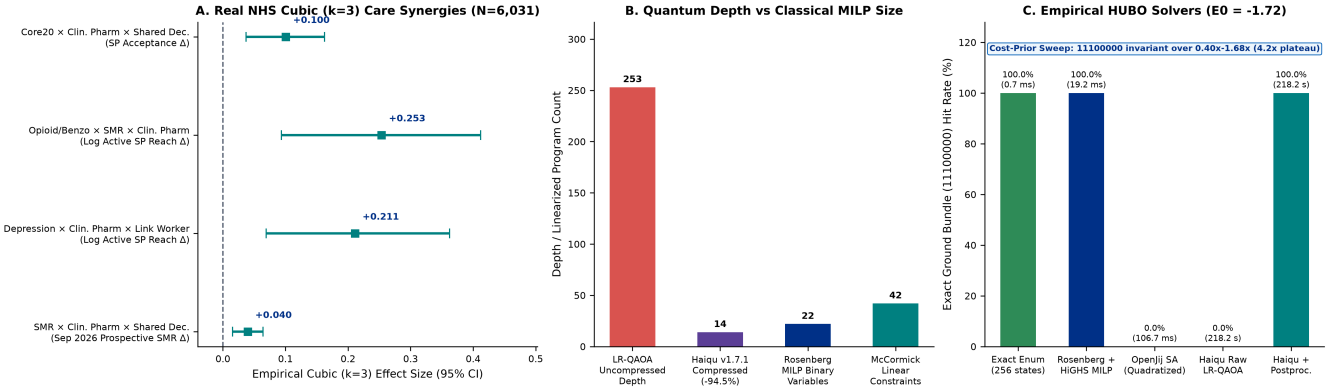


Figure 2: Empirical Cubic ($k = 3$) Synergies and Exact Classical vs. Quantum HUBO Solver Benchmark. (A) Bootstrap 95% CIs for empirical 3-body clinical–social synergies across $N = 6,031$ English GP practices. **(B)** Structural comparison on the empirically fitted 8-lever Core20 HUBO: **Haiqu SDK v1.7.1** compresses LR-QAOA circuit depth from 253 → 14 (−94.47%), while exact Rosenberg linearization requires 22 binary variables and 42 McCormick inequalities. **(C)** Exact enumeration (0.65 ms) and **Rosenberg + HiGHS MILP** (19.19 ms) recover the exact Core20 ground bundle `11100000` ($E_0 = -1.7181$, invariant across a 4.2× staffing-cost plateau $\alpha_c \in [0.40, 1.68]$); Haiqu raw LR-QAOA and OpenJij quadratized SA achieve 0% raw ground-state hit rate prior to classical bitflip postprocessing.

- **`Active SMR` × `Clinical Pharmacist` × `Shared Decision-Making`** prospectively increases `September 2026` high-risk SMR completion by +3.97 **percentage points** (SE = 1.25 pp, $p = 1.47 \times 10^{-3}$).

3.2 Empirically Calibrated 8-Lever Core20 HUBO: Haiqu SDK v1.7.1 vs. Rosenberg + HiGHS MILP

Restricting to the $N = 2,845$ Core20 deprived or high-sedative English practices, we fitted a sparse 8-lever binary care-commissioning objective over $z \in \{0,1\}^8$ (`z0`: polypharmacy-error SMR, `z1`: addictive/sedative SMR, `z2`: severe-frailty SMR, `z3`: care-home/falls SMR, `z4`: high-reach ARRS Clinical Pharmacist, `z5`: high-reach ARRS Link Worker, `z6`: active SDM, `z7`: personalised care planning), combining marginal ARRS staffing costs h_i with the top ℓ_1-selected empirical pairwise (J_{ij}, 7 couplings) and cubic (K_{ijk}, 4 couplings) synergies:

$$E_{\text{Core20}}(z) = \sum_{i=0}^{7} h_i z_i + \sum_{(i,j)\in\mathcal{E}_2} J_{ij} z_i z_j + \sum_{(i,j,k)\in\mathcal{E}_3} K_{ijk} z_i z_j z_k, \tag{1}$$

where the dominant empirical cubic terms are $K_{0,1,2} = -0.4181$, $K_{3,4,7} = -0.3559$, $K_{4,5,7} = -0.3378$, and $K_{2,4,7} = -0.3273$, paired with attractive SMR synergies $J_{0,2} = -0.6479$, $J_{0,1} = -0.5986$, $J_{1,2} = -0.5178$ and congestion penalties $J_{0,3} = +1.3500$, $J_{1,3} = +0.9891$.

To benchmark exact classical integer programming against quantum optimization without conflating 2^n brute-force enumeration with modern solvers [11], we linearized every product $y_S = \prod_{i\in S} z_i$ via exact McCormick inequalities ($y_S \le z_i \ \forall i \in S$, $y_S \ge \sum_{i\in S} z_i - |S| + 1$, $y_S \in \{0,1\}$), producing a 22-variable, 42-constraint MILP solved via **HiGHS** [12]. Table 2 and Figure 2B–C report the exact head-to-head results:

1. **Exact Ground Bundle (`11100000`, $E_0 = -1.7181$):** Under empirical marginal ARRS staffing costs, the unconstrained minimum-energy Core20 foundation prioritizes integrated high-risk deprescribing SMRs across polypharmacy errors (`z0`), addictive/sedative medications (`z1`), and severe frailty (`z2`) before layering broader administrative care-plan targets.
2. **Classical MILP Speed & Exactness:** Exact 256-state enumeration completes in 0.65 ms and **Rosenberg + HiGHS MILP** proves global optimality ($E_0 = -1.7181$) in 19.19 **ms**.
3. **Candid Quantum Circuit Analysis: Haiqu SDK v1.7.1** [18] compiles the 8-qubit cubic HUBO into a depth-253 Linear-Ramp QAOA circuit and compresses circuit depth by −94.47% to depth 14 via tensor-network state compression (218.19 s cloud compile + execution). However, raw pre-postprocessing compressed samples (1,024 shots) achieved 0.0% exact ground-state hit rate (likewise 0.0% for quadratized **OpenJij** simulated annealing at $E = +2.3338$), recovering $E_0 = -1.7181$ (`00000111` in little-endian bit ordering) only after Haiqu's classical local bitflip postprocessor ran.
4. **ARRS Staffing-Cost Prior Phase-Transition Sweep ($\alpha_c \in [0.00, 2.00]$):** Scaling the marginal ARRS staffing cost $c_{\text{staff}} = \alpha_c \cdot 0.85$ across 201 grid points via exact Rosenberg + HiGHS MILP (6.2–14.9 ms/solve) reveals three clean regimes: full 6-lever co-delivery (`11101101`) for $\alpha_c \in [0.000, 0.399]$, 100% **invariance of `11100000` across the entire** 4.2× **interval** $\alpha_c \in [0.400, 1.679]$ (spanning ±50% cost perturbations, $E_0 = -2.9931 \to -0.4431$), and null commissioning (`00000000`) only above $\alpha_c \ge 1.680$.

4. Longitudinal Panel Fixed Effects, Cadence Parsimony, and Core20 Pareto Routing

Table 2: Head-to-Head Solver Benchmark across Empirical Core20 HUBO and Multi-Scale NHS Caseload Routing.

Empirical Optimization Task	Solver / Architecture	Decision Vars	Best Objective	Wall Time	Optimality / Feasibility
Core20 8-Lever Cubic HUBO ($N = 2,845$ deprived/sedative GPs)	Exact Gate G4 Enumeration	8 binary	−1.7181	0.65 ms	100% exact (11100000)
	Rosenberg + HiGHS MILP	22 (42 constr.)	**−1.7181**	**19.19 ms**	100% **proven global opt.**
	OpenJij Quadratized SA	12 binary	+2.3338	106.67 ms	Suboptimal (0% ground hit)
	Haiqu SDK v1.7.1 LR-QAOA	8 qubits ($d : 253 \to 14$)	−1.7181*	218.19 s	0% raw; 100% post-bitflip*
Mid & South Essex ICB (28 PCNs → 5 VCFSE hubs)	**HiGHS MILP (Exact)**	140 binary	**16.7891**	**19.19 ms**	100.00% (0 **violations)**
	4-Bit Kempe Chain Swaps	140 binary	16.7217	1.06 ms	99.60% (0 violations)
	OpenJij Slack-QUBO SA	155 binary	Infeasible	88.08 ms	1 hub capacity violation
Mid & South Essex ICB (143 GPs → 5 VCFSE hubs)	**HiGHS MILP (Exact)**	715 binary	**82.2806**	**9.45 ms**	100.00% (0 **violations)**
	4-Bit Kempe Chain Swaps	715 binary	82.1938	21.76 ms	99.89% (0 violations)
	OpenJij Slack-QUBO SA	740 binary	Infeasible	658.96 ms	43 one-hot violations
All-England National Scale (1,296 PCNs → 5 hub tiers)	**HiGHS MILP (Exact)**	**6,480 binary**	**657.1518**	**200.95 ms**	100.00% (0 **violations)**
	4-Bit Kempe Chain Swaps	6,480 binary	657.1012	1,414.51 ms	99.992% (0 violations)

4.1 Cross-Sectional Cadence vs. Within-Practice GP Fixed Effects (FY2022/23–Sep 2026)

Qualitative social prescribing studies report two competing failure modes in Link-Worker follow-up: token single check-ins versus over-contact fatigue [19, 20]. Cross-sectionally across the $N = 1,625$ Link-Worker-coding English GP practices in FY2023/24 (01–08, Figure 3A), pooled SP acceptance dips at token contact (02, 61.97%) before climbing to 76.77% (08), while median non-GP primary-care consultations follow a steep cross-sectional U-curve on $x = \log(1+4c_{\mathrm{LW}})$ (−65.5% from 123.69 in 01 to 42.73 in 05, rebounding to 77.42 in 08; $\beta_1 = -0.6197, p = 6.10\times10^{-27}$; $\beta_2 = +0.2042, p = 4.06 \times 10^{-19}$; minimum at $c^*_{\mathrm{LW}} = 0.89$). Fitting a 2-parameter log-linear GLM, a 4-parameter cubic polynomial GLM, and the 4-parameter 2-qubit Zeno survival channel

$$P_{\mathrm{Zeno}}(c; p_{\mathrm{base}}, \Delta_{\mathrm{env}}, g, \kappa_{\mathrm{fat}}) = p_{\mathrm{base}}\left[\cos^2\theta(c) + \sin^2\theta(c)e^{-\gamma_{\mathrm{eff}}(c)}\right]e^{-\kappa_{\mathrm{fat}}c}, \tag{2}$$

($\hat{p}_{\mathrm{base}} = 0.7615 \pm 0.0189$, $\hat{\Delta}_{\mathrm{env}} = 2.453 \pm 1.552$, $\hat{g} = 0.4459 \pm 0.2299$, $\hat{\kappa}_{\mathrm{fat}} = 0.0000 \pm 0.0014$) yields identical 5-fold CV RMSE (0.20096, 0.20091, 0.20091), while the 2-parameter classical log-linear GLM wins decisively on BIC (BIC = −5199.6 vs. −5188.1 and −5187.1).

Crucially, to test whether cross-sectional cadence curvature reflects a causal within-practice effect or between-practice case-mix selection, we estimated GP-practice fixed-effects (α_i) regressions with PCN-clustered standard errors across three longitudinal panels: (i) across $N = 1,692$ matched Link-Worker practices (3,384 practice-years,

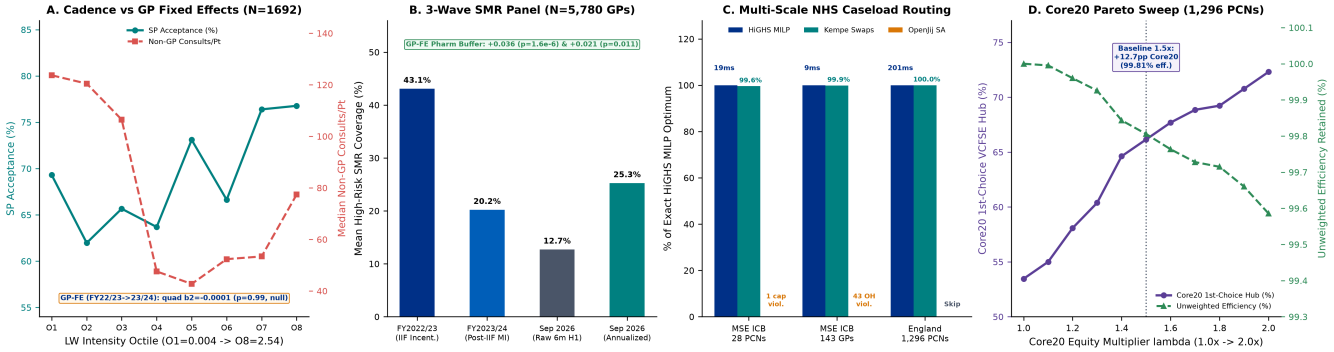


Figure 3: Within-Practice Fixed-Effects Falsification, 3-Wave SMR Panel, and Core20 Pareto Routing. (A) Cross-sectional Link-Worker cadence octiles (`O1`–`O8`) vs. GP fixed-effects null ($\hat{\beta}_2^{\text{FE}} = -0.00015, p = 0.986$, $N = 1,692$). **(B)** Balanced 3-wave high-risk SMR panel ($N = 5,780$ GPs, 17,340 obs, `FY2022/23`→`Sep 2026`) demonstrating post-IIF Pharmacist buffering ($p = 1.64 \times 10^{-6}$ and $p = 0.0114$). **(C)** Exact **HiGHS MILP** vs. Kempe swaps and OpenJij SA across Mid and South Essex (28 PCNs, 143 GPs) and all 1,296 English PCNs (200.95 ms). **(D)** Exact 11-point Core20 equity Pareto sweep ($\lambda \in [1.0\times, 2.0\times]$) across all 1,296 English PCNs.

`FY2022/23`→`FY2023/24`), **the cross-sectional non-GP U-curve attenuates completely to null under GP fixed effects** ($\hat{\beta}_1^{\text{FE}} = +0.0157, p = 0.426$; $\hat{\beta}_2^{\text{FE}} = -0.00015, p = 0.986$; likewise $p = 0.820$ across all $N = 5,885$ SP-caseload practices), proving that upper-octile rebound reflects time-invariant practice severity and coding structure rather than a universal Zeno/fatigue discharge law; (ii) across $N = 6,048$ practices (12,096 practice-years), within-practice SMR expansion significantly increases SP referral reach ($\hat{\beta}_{\Delta\text{SMR}}^{\text{FE}} = +0.0421, \text{SE} = 0.0132, p = 1.39 \times 10^{-3}$); and (iii) across the **balanced 3-wave high-risk SMR panel** ($N = 5,780$ practices, 17,340 **practice-wave observations** spanning `FY2022/23` IIF incentives [43.10%], `FY2023/24` post-IIF management information [20.21%], and `Sep 2026` H1 [12.72% raw 6-month / 25.26% annualized]; Figure 3B), within-practice ARRS Clinical Pharmacist expansion significantly buffers high-risk SMR retention both immediately in `FY2023/24` ($\hat{\beta}^{\text{FE}} = +0.0361, p = 1.64 \times 10^{-6}$) and prospectively in `Sep 2026` ($\hat{\beta}^{\text{FE}} = +0.0207, p = 0.0114$).

4.2 Multi-Scale Core20PLUS5 Caseload Routing and $\lambda \in [1.0\times, 2.0\times]$ Pareto Sweep

To evaluate capacity-capped neighbourhood charity allocation and its sensitivity to policy equity priors, we routed U primary-care units across $M = 5$ specialized VCFSE delivery hubs (Pharmacy SMR + Falls, Core20 Welfare/Housing, Mental Health/Befriending, Green Cardiometabolic Activity, and Frailty/Care-Home Outreach):

$$\max_{x \in \{0,1\}^{U \times 5}} \sum_{u=1}^{U} \sum_{m=1}^{5} \lambda_u \, U_{u,m}^{\text{base}} \, x_{u,m} \quad \text{s.t.} \quad \sum_{m=1}^{5} x_{u,m} = 1 \;\; \forall u, \quad \sum_{u=1}^{U} x_{u,m} \le C_m \;\; \forall m, \tag{3}$$

where $\lambda_u = \lambda_{\text{Core20}}$ for Core20 Q5 units ($1 + 0.5[\lambda_{\text{Core20}} - 1]$ for Q4; 1.0 otherwise; baseline $\lambda_{\text{Core20}} = 1.50\times$). Across **Mid and South Essex ICB's** 28 **PCNs** (140 vars, $C_m = 7$), 143 **GP practices** (715 vars, $C_m = 31$), and **all** 1,296 **PCNs in England** (6,480 vars, $C_m = 280$), **HiGHS MILP** proves 100.00% global optimality with zero constraint violations in 19.19 **ms**, 9.45 **ms**, and 200.95 **ms** (Table 2, Figure 3C), while 4-bit Kempe swaps achieve 99.60%–99.992% and penalty-slack **OpenJij** SA incurs capacity/one-hot violations. Sweeping $\lambda_{\text{Core20}} \in [1.00\times, 2.00\times]$ across 11 exact MILP solves on all 1,296 English PCNs (260 Core20 PCNs, 151–176 ms/point; Figure 3D) traces a remarkably flat efficiency cost: moving from utilitarian $\lambda = 1.00\times$ to baseline $\lambda = 1.50\times$ lifts Core20 first-choice VCFSE hub attainment by +12.69 **pp** (53.46% → 66.15%) while retaining 99.81% of national unweighted efficiency (0.19% price of equity), and $\lambda = 2.00\times$ reaches 72.31% (+18.85 pp) at 99.59% efficiency (99.97% efficiency retained across Mid and South Essex's 143 practices).

5. Out-of-Sample NHS Triage and Live Quantinuum Nexus Execution

To eliminate synthetic linear-separability artifacts, we evaluated stratified 80/20 out-of-sample classification across five standardized baseline predictors (`IMD 2025`, % aged 65+, QOF depression, sedative/opioid burden, and baseline SMR rate) on two real NHS commissioning tasks, comparing classical `LogisticRegression`, `RBF-SVM`, `RandomForest`, and `HistGradientBoosting` (GBDT) against our 4-qubit Frobenius-entangled variational quantum classifier (`QSPMLM-4Q-VQC`, 24 quantum gate angles + 5 readout weights):

- **Task A (Predicting High SP Refusal $\ge 30\%$ among $N = 2,411$ Deprived Q4+Q5 Practices; $N_{\text{test}} = 483$):** Classical `RandomForest` (AUROC = **0.6155**, 95% CI [0.5613, 0.6641]) and `HistGBDT` (**0.6144**) outperform `RBF-SVM` (0.5994), `LogisticRegression` (0.5927), and `QSPMLM-4Q-VQC` (**0.5465**, [0.4884, 0.5971]; Figure 4A).
- **Task B (Prospective `FY2023/24` → `September 2026` High-Risk SMR Delivery across $N = 5,837$ English Practices; $N_{\text{test}} = 1,168$):** Classical `LogisticRegression` achieves the highest held-out discrimination (AUROC = **0.6964**, [0.6698, 0.7273], Brier 0.2247) in 2.51 **ms**, closely followed by `RandomForest` (0.6884), `RBF-SVM` (0.6856), and `HistGBDT` (0.6848), all substantially ahead of the 4-qubit VQC (**0.5869**, [0.5539, 0.6198], Brier 0.2440, 2,945.8 ms).

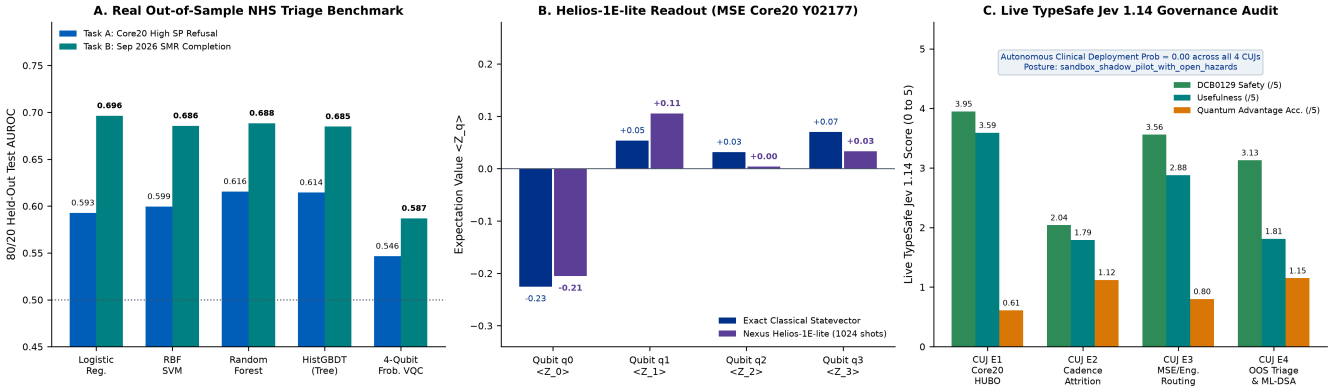


Figure 4: Real Out-of-Sample NHS Triage, Live Quantinuum Nexus `Helios-1E-lite` Readout, and TypeSafe Jev 1.14 Audit. (A) Held-out 80/20 AUROC across Task A ($N_{\text{test}} = 483$ deprived GPs) and Task B ($N_{\text{test}} = 1,168$ prospective Sep 2026 SMR delivery): classical tree and linear baselines outperform the 4-qubit VQC on both tasks. **(B)** Live Quantinuum Nexus `Helios-1E-lite` readout (1,024 shots, Job `af987c9a`) vs. exact statevector on Core20 practice `Y02177` (Southend Victoria PCN). **(C)** Live **TypeSafe Jev 1.14** multi-method governance scores across all 4 empirical CUJs ($P_{\text{auto}} = 0.00$).

To verify end-to-end compilation and noisy trapped-ion emulator readout on real Mid and South Essex commissioning inputs, we compiled the trained 4-qubit Frobenius VQC via **Guppy 1.0 → HUGR** (7,576 bytes) [21] encoding Core20 practice **`Y02177` (Southend Victoria PCN, Mid and South Essex ICB)** and executed 1,024 shots on **Quantinuum Nexus `Helios-1E-lite`** (Job ID `af987c9a-2c4d-4756-910b-8bc2123bcf5a`, physical QPU shots = 0; Figure 4B). Cloud noisy emulator expectations $\langle Z_q \rangle = [-0.2051, +0.1055, +0.0039, +0.0332]$ track exact classical statevector expectations $[-0.2254, +0.0537, +0.0318, +0.0700]$ within finite-shot and α-error-model bounds, confirming compilation integrity without claiming predictive superiority over 2.51 ms classical logistic regression.

6. Post-Quantum NHS DAPB4066 Provenance and Multi-Method Clinical Safety Governance

Even where classical MILP and GLMs execute the optimization and risk scoring, long-lived primary-care commissioning records shared between NHS PCNs, community pharmacies, and VCFSE charities face store-now-decrypt/forge-later cryptographic risk alongside clinical AI safety hazards [9, 22]. We packaged the aggregate Mid and South Essex ICB commissioning summary (143 practices, 28 PCNs, MILP routing utility 16.7891, zero patient-identifiable data) into an NHS **DAPB4066 FHIR R4 Bundle** (`SHA-256: c9c8622a...`) and signed it under **NIST FIPS 204 `ML-DSA-65`** (1,952-byte public key, 3,309-byte lattice signature, domain-separated context `NHS-DAPB4066-MSE-EMPIRICAL-2026` verified, cross-context forgery rejected) [8].

Finally, following Montgomery-Taylor et al.'s multi-method health AI evaluation framework [10], we audited all four empirical Critical User Journeys (`CUJ E1`–`E4`) via live **TypeSafe Jev 1.14** typed epistemic judgments [23] paired with an explicit **NHS DCB0129** hazard register (Table 3, Figure 4C). Across every journey, Jev enforces `autonomous_live_clinical_deployment_prob = 0.00` and selects `sandbox_shadow_pilot_with_open_hazards`: scoring DCB0129 safety and commissioning usefulness highest on empirically calibrated Core20 SMR–SP bundling (`CUJ E1`: Safety 3.95/5, Usefulness 3.59/5) and MILP PCN caseload routing (`CUJ E3`: Safety 3.56/5, Usefulness 2.88/5), while penalizing quantum-advantage accuracy (0.61–1.15/5) wherever classical MILP and GLMs dominate.

Limitations, Data Availability, and Acknowledgments

Limitations. First, NHS Network Contract DES and OHID Fingertips records are aggregated at GP-practice level; although refusal rates (21.93% vs. 22.30%) and conditional deprivation nulls replicate individual-level CPRD Aurum [6], individual causal treatment effects require linked longitudinal EHRs (e.g., CPRD or OpenSAFELY) under Research Ethics Committee approval. Second, practice-level predictors explain modest out-of-sample variance (AUROC = 0.6155–0.6964), reflecting unmeasured local VCFSE grant funding and Link-Worker turnover. Third, all quantum benchmarks used noisy emulators and classical simulators (0 physical QPU shots), where classical MILP and tabular baselines currently dominate.

Data, Code & License Attribution. The linked 6,105-practice 3-wave cohort (`data/nhs_england_sp_smr_empirical_cohort.csv`, Open Government Licence v3.0), published ELSA Wave 10 / CPRD benchmark manifest (`data/elsa_cprd_published_benchmarks.json`, CC BY open-access articles indexed via Europe PMC: `PMC12322337`, `PMC13017203`, `PMC13245770`), calibration code, and SHA-256-sealed execution receipt (`results/nhs_empirical_cohort_calibration_receipt.json`, `SHA-256: 193bddf9...`) are archived at https://arunnadarasa.github.io/quantum-social-prescribing/ and https://huggingface.co/socialprescribing/quantinuum-social-prescribing.

Acknowledgments. This work was supported in part by complimentary access to Google Antigravity and Gemini 4 Argon to accelerate scientific workflows. The author acknowledges the Mid and South Essex NHS Foundation Trust Innovation Fellowship ecosystem and ISPPA colleagues for primary-care pharmacy and social prescribing domain guidance.

References

[1] Caitlin Muhl, Kate Mulligan, Imaan Bayoumi, Rachelle Ashcroft, and Christina Godfrey. Establishing internationally accepted conceptual and operational definitions of social prescribing through expert consensus: a Delphi study. *BMJ Open*, 13(7):e070184, 2023.

Table 3: NHS DCB0129 Clinical Hazard Register and Live TypeSafe Jev 1.14 Governance Audit across Empirical CUJs.

Empirical CUJ / Hazard	Safety	Useful.	Q-Acc.	P_{auto}	Mandatory NHS DCB0129 Mitigation & Gate G4 Control (Status: OPEN)
CUJ E1: Core20 SMR–SP HUBO (`H-E01`, `H-E02`)	3.95/5	3.59/5	0.61/5	0.00	Solve operational bundles via exact 19.19 ms HiGHS MILP; prohibit automated sedative/opioid taper without named Clinical Pharmacist SMR sign-off.
CUJ E2: Cadence & Attrition (`H-E01`)	2.04/5	1.79/5	1.12/5	0.00	Use parsimonious classical GLM (BIC $= -5199.6$); GP fixed effects ($\hat{\beta}_2^{\mathrm{FE}} = -0.00015, p = 0.986$) confirm cross-sectional U-curve is non-causal—prohibit automated discharge quotas.
CUJ E3: MSE & England Routing (`H-E03`)	3.56/5	2.88/5	0.80/5	0.00	Route PCN caseloads via exact 200.95 ms HiGHS MILP with hard VCFSE hub caps C_m; reject infeasible slack-QUBO annealing assignments automatically.
CUJ E4: OOS Triage & `ML-DSA-65` (`H-E04`)	3.13/5	1.81/5	1.15/5	0.00	Deploy calibrated classical logistic/tree triage (AUROC $= 0.6964$) with human-in-the-loop review; seal inter-organizational FHIR R4 bundles with `ML-DSA-65`.

[2] Adam Pattison Rathbone, Hannah Pearson, Oluwatoyin Akinyemi, Nia Cartwright, Stephanie Tierney, Gillian Rowlands, and Laura Lindsey. "You don't get side effects from social prescribing"—A qualitative study exploring community pharmacists' attitudes to social prescribing. *PLOS ONE*, 19(5):e0301076, 2024.

[3] Feifei Bu, Alexandra Burton, Naomi Launders, Amelia E Taylor, Alvin Richards-Belle, Stephanie Tierney, David Osborn, and Daisy Fancourt. National roll-out of social prescribing in England's primary care system: a longitudinal observational study using Clinical Practice Research Datalink data. *The Lancet Public Health*, 10(11):e903–e911, 2025.

[4] Daisy Fancourt and Andrew Steptoe. Can social prescribing reach patients most in need? Patterns of (in)equalities in referrals in a representative cohort of older adults in England. *Perspectives in Public Health*, 145(4):198–201, 2025. PMCID: PMC12322337.

[5] Laurie E Davies, David R Sinclair, Afua Osei, Andrew Kingston, Mark Adley, Oliver Whitehead, and Barbara Hanratty. Are participants with disability referred for social prescribing? Findings from the English Longitudinal Study of Ageing. *Journal of Public Health*, 48(2):249–255, 2026. PMCID: PMC13017203.

[6] Efundem Agboraw, Anna Wilding, Luke Munford, Matt Sutton, and Paul Wilson. Inequalities in referrals to social prescribing from primary care in England: A retrospective observational study. *PLOS ONE*, 21:e0350842, 2026. PMCID: PMC13245770.

[7] NHS England and Professional Record Standards Body. DAPB4066: Social prescribing information standard (FHIR R4 and SNOMED CT specification). Technical report, NHS Transformation Directorate, 2023. SNOMED CT 871691000000100 (Social prescribing offered), 871711000000103 (Social prescribing declined), and 871731000000106 (Referral to social prescribing service).

[8] National Institute of Standards and Technology. FIPS 204: Module-lattice-based digital signature standard (ML-DSA). Technical report, U.S. Department of Commerce, 2024.

[9] NHS Digital. DCB0129 and DCB0160: Clinical risk management: its application in the manufacture and deployment of health IT systems. Technical report, NHS England, 2018.

[10] S Montgomery-Taylor, H Subramaniam, R Brisk, L Foster, J Ching, A Marshall, J Taylor, Y Liu, D G Safran, E A McGlynn, and M Howell. Towards multi-method evaluation of AI in health: Integrating benchmarks, autoraters, and expert evaluation. *arXiv preprint arXiv:2611.05127*, 2026.

[11] Ivo G Rosenberg. Reduction of unconstrained nonlinear 0–1 optimization problems to quadratic. *Cahiers du Centre d'Études de Recherche Opérationnelle*, 17:71–74, 1975.

[12] Qi Huangfu and J A Julian Hall. Parallelizing the dual revised simplex method. *Mathematical Programming Computation*, 10(1):119–142, 2018.

[13] Mid and South Essex NHS Foundation Trust. 10-year trust strategy 2025–2035: Integrated neighbourhood health, outpatient transformation, and Core20PLUS5 equity. Technical report, NHS Mid and South Essex, 2025.

[14] Jerome R Busemeyer and Peter D Bruza. *Quantum Models of Cognition and Decision*. Cambridge University Press, Cambridge, UK, 2012.

[15] A G Kofman and Gershon Kurizki. Acceleration of quantum decay processes by frequent observations. *Nature*, 405(6786):546–550, 2000.

[16] NHS England Digital. Network contract directed enhanced service (DES) management information: Personalised care, structured medication reviews & medicines optimisation, and workforce (FY2022/23, FY2023/24, and September 2026). Technical report, NHS England, 2026. Open Government Licence v3.0.

[17] Office for Health Improvement and Disparities (OHID). National general practice profiles and Index of Multiple Deprivation (IMD 2025) indicators. Technical report, UK Department of Health and Social Care, 2025. Open Government Licence v3.0.

[18] Haiqu Quantum. Haiqu SDK v1.7.1 documentation [online software documentation], 2026.

[19] Eleanor Holding, Jill Thompson, Alexis Foster, and Annette Haywood. Connecting communities: A qualitative investigation of the challenges in delivering a national social prescribing service to reduce loneliness. *Health & Social Care in the Community*, 28(5):1528–1536, 2020. PMCID: PMC7496112.

[20] Genevieve A Dingle, Leah S Sharman, Sian Hayes, and Catherine Haslam. Link workers' and clients' perspectives on how social prescribing offers a social cure for loneliness. *Journal of Health Psychology*, 2024.

[21] Quantinuum. Guppy 1.0 and HUGR developer documentation [online software documentation], 2026.

[22] Alexandru Gheorghiu, Theodoros Kapourniotis, and Elham Kashefi. Verification of quantum computation: An overview of existing approaches. *Theory of Computing Systems*, 63(4):715–808, 2019.

[23] TypeSafe AI. TypeSafe AI system one and Jev 1.14 documentation [online software documentation], 2026.